

Gabrielle Johnson's Feasibility Memo

The proposed dataset appears insufficiently viable for training an AI assisted triage model at Mercer's ED.

Dataset Summary

The dataset contains 55,121 emergency department patient encounters across 225 columns, approximately 200 of which are chief complaint indicators. The dataset is adult only, as none of the patients are under 18 years old, while approximately 32% are over the age of 65. In terms of demographics, women are the majority of the patients recorded at 58%, additionally, 80% of patients are non-hispanic with 53% being White/Caucasian.

Triage acuity was recorded using the Emergency Severity Index (ESI), which ranges from 1 to 5, where 1 represents the most urgent cases and 5 represents the least urgent cases. Nearly half (49%) of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1. The most common chief complaint was abdominal pain, representing 12.8% of encounters, followed by "other" at 7.8%. The dataset also includes routinely collected triage vital signs, including heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose.

Top 3 Quality Concerns

1. Missingness and incomplete safety variables

Some important triage and safety variables are missing or unavailable. Pain score is not present, even though it would be clinically useful for triage decision support. Pregnancy status is also unavailable, which limits the ability to test the proposed Mercer safety guardrails for pregnant patients. Missing vital signs were handled through imputation, and missingness flags were added where appropriate, but these gaps still limit how fully the dataset can reflect Mercer's triaging.

2. Triage Level Imbalance

The dataset is heavily concentrated around mid acuity patients, with nearly half of all encounters classified as ESI 3 and fewer than 1% classified as ESI 1. This creates a concern because the model may learn the common ESI categories better than the rare but most urgent cases. This is severely alarming as the model may miss the sickest patients. ESI specific performance, especially for ESI 1, should be tested separately rather than relying on overall accuracy.

3. Representativeness and transferability risk

The dataset is adult only, with no patients under 18, and the racial distribution is majority White/Caucasian. This may not reflect Mercer General ED's Caribbean patient population, where demographic and chronic illness patterns will differ. Because of this, the dataset should only be treated as an exploratory baseline and should not be used without local validation.

Top 3 Reasons Not to Proceed

1. The dataset does not fully represent Mercer's target population

The dataset is adult only, with no patients under 18, and the racial distribution is majority White/Caucasian. This limits transferability as Mercer serves a Caribbean population with a contrasting racial profile, mixed age population and experiences different environmental, and disease pattern contexts. Tropical conditions, local chronic disease, and race/ethnicity may affect chief complaints, acuity, and triage decisions.

2. Important proposed triage variables are missing and safety guardrails

Pain score is not available in the dataset, even though it would be clinically useful for triage decision support. Pregnancy status is also unavailable, which limits the ability to build the safety checks proposed for pregnant patients. This is important because the proposed Mercer system includes additional safeguards for paediatric, elderly and pregnant patient groups.

3. Severe class imbalance may weaken high acuity prediction and vague chief complaint category may weaken clinical interpretability

Nearly half of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1. This creates a class imbalance concern because a model may achieve high overall accuracy by predicting common ESI categories while performing poorly on rare but clinically critical ESI 1 cases. Since ESI 1 patients represent the most urgent cases it's crucially important for the model to be able to detect high acuity patients.

The most common chief complaint was abdominal pain at 12.8%, followed by "other" at 7.8%. Because "other" is the second largest chief complaint category, this may hide clinically decisive symptoms inside one vague label. This may weaken model training because many different symptoms are grouped under "other," making it harder for the model to learn clear chief complaint patterns.

Caveats

The dataset may still be useful for exploratory analysis, including reviewing vital sign patterns, chief complaint relationships, and ESI distributions, but it is not sufficient for deployment without local validation. Paediatric safety and Caribbean population generalizability remain unresolved because the dataset is adult only and majority White/Caucasian. Pain score and pregnancy status are missing, which limits the ability to test the proposed Mercer safety guardrails. The “other” chief complaint category is the second most common complaint, reducing clinical specificity and potentially weakening the model’s ability to learn clear chief complaint patterns. Finally, ESI 1 cases make up less than 1% of encounters, so the model may fail to identify the most urgent cases unless high acuity performance is tested separately.